

Load Balancer

Praktikum Cloud Computing

Tim Pengajar MK. Komputasi Awan

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Praktikum 3

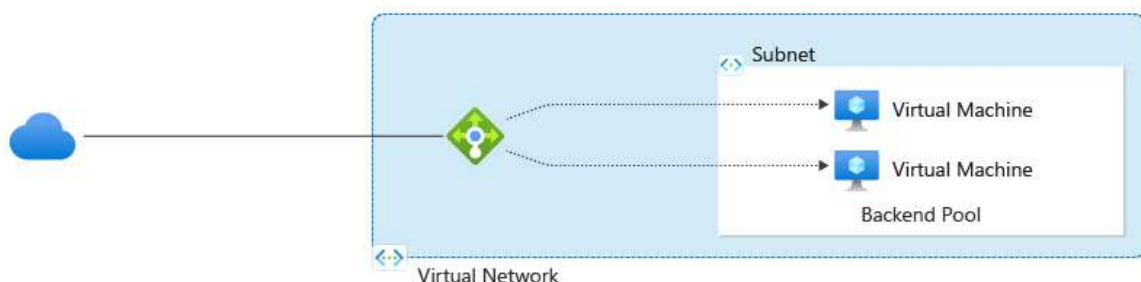
Load Balancer Menggunakan Microsoft Azure

A. Azure Load Balancer

Load balancing adalah proses pendistribusian beban terhadap sebuah servis yang ada pada sekumpulan server atau perangkat jaringan ketika ada permintaan dari pemakai. Ketika banyak permintaan dari pemakai maka server tersebut akan terbebani karena harus melakukan proses pelayanan terhadap permintaan pemakai. Solusi nya adalah dengan membagi – bagi beban yang datang ke beberapa server, jadi tidak berpusat ke salah satu perangkat jaringan saja.

Azure Load Balancer adalah layanan cloud yang mendistribusikan lalu lintas jaringan masuk ke mesin virtual (VM) atau set skala mesin virtual (VMSS) di backend. Azure Load Balancer beroperasi pada lapisan 4 dari model Open Systems Interconnection (OSI). Ini adalah titik kontak tunggal untuk klien. Layanan ini mendistribusikan lalu lintas masuk yang tiba di frontend load balancer ke instance kumpulan backend. Lalu lintas data ini didistribusikan sesuai dengan aturan penyeimbangan beban dan pemantauan kesehatan yang dikonfigurasi. Instans kumpulan backend dapat berupa mesin virtual (VM) Azure atau kumpulan skala mesin virtual. Penyeimbang beban publik dapat menyediakan konektivitas masuk dan keluar untuk mesin virtual di dalam jaringan virtual pengguna. Untuk skenario lalu lintas masuk, Azure Load Balancer dapat menyeimbangkan lalu lintas internet ke VM pengguna. Untuk skenario lalu lintas keluar, layanan dapat menerjemahkan alamat IP privat VM ke alamat IP publik untuk koneksi keluar apa pun yang berasal dari VM pengguna.

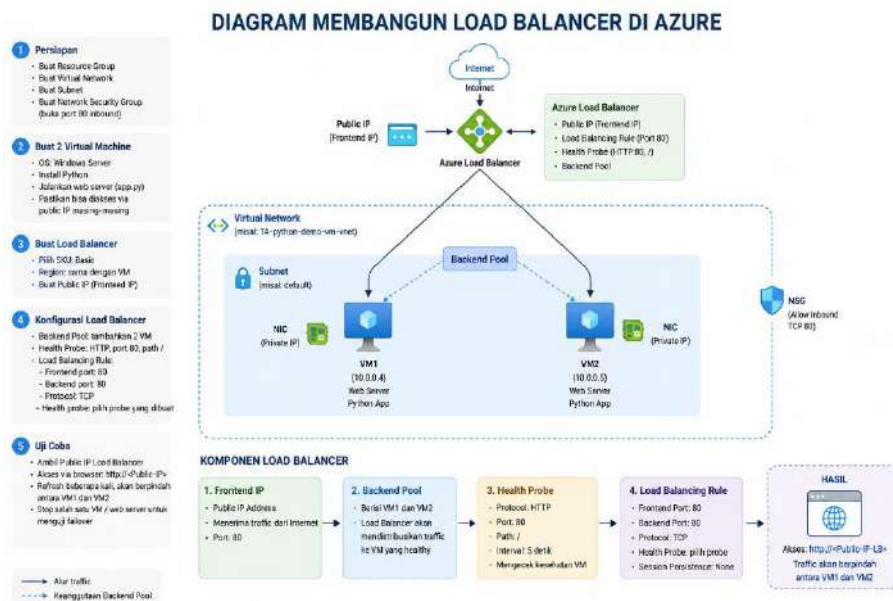
Sebagai alternatif, load balancer digunakan untuk menyeimbangkan lalu lintas di dalam suatu jaringan virtual. Dengan load balancer internal, pengguna dapat menyediakan konektivitas masuk ke VM dalam skenario konektivitas jaringan privat, seperti mengakses frontend load balancer dari jaringan lokal dalam skenario hibrid.



Gambar 1. Ilustrasi Azure Load Balancer

B. Komponen Azure Load Balancer

Dalam membangun Load Balancer di dalam Azure berikut tahapan yang harus dilakukan: yaitu:



Gambar 2. Tahapan membangun Load Balancer di Azure

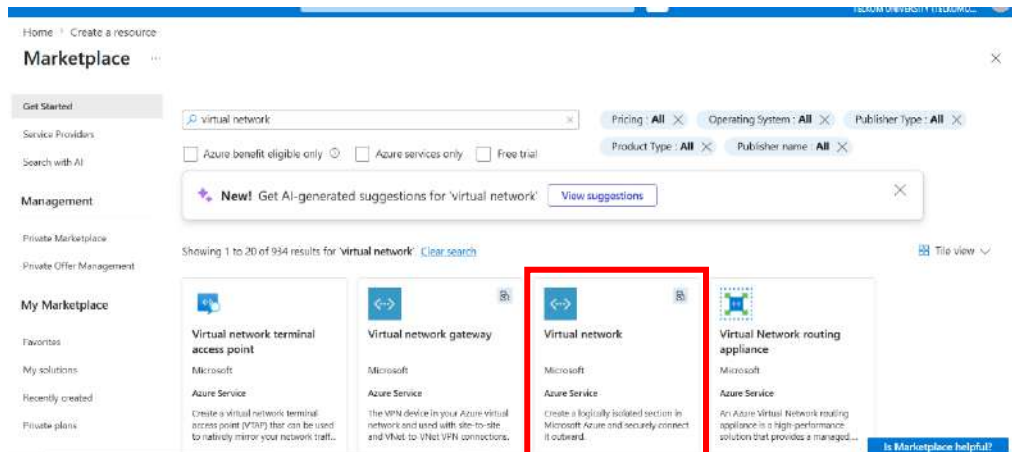
1. Membuat *Resource Group* khusus untuk sistem yang akan diimplementasikan Load Balancer
2. Membuat *Virtual Network* untuk konfigurasi IP dan Subnet
3. Membuat *Virtual Machine* sebagai *Backend Service*
4. Membuat *Load Balancer* yang akan digunakan untuk mengatur lalu lintas jaringan
5. Menguji *Load Balancer* yang sudah dibuat

C. Tahapan Membangun Load Balancer

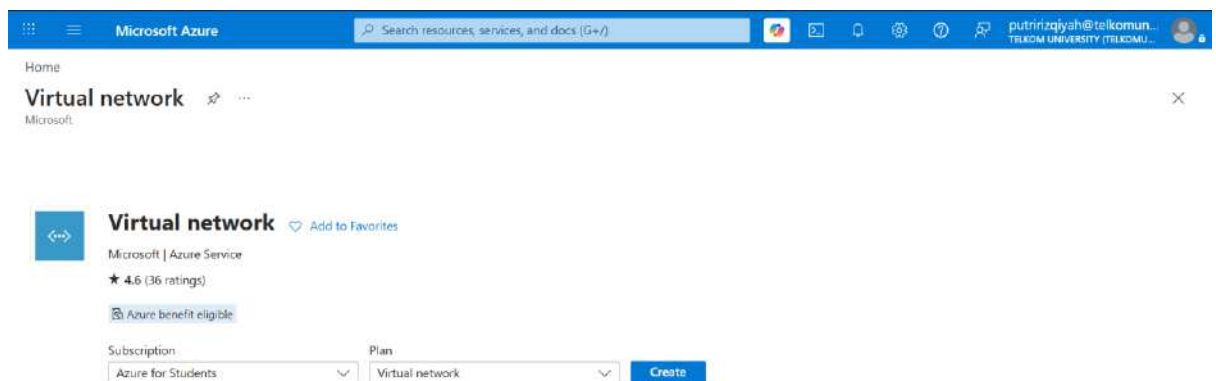
Untuk membangun Load Balancer di dalam Azure berikut langkah-langkah yang harus dilakukan

A. Membuat Virtual Network

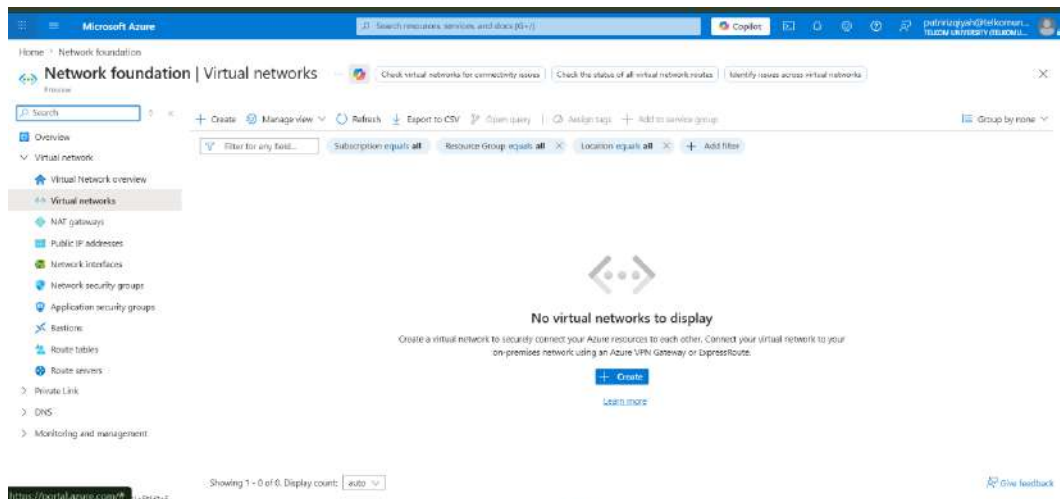
1. Tahapan yang pertama yaitu membangun *Virtual Network*, lakukan pencarian di *search bar* dan pilih menu *Virtual Network*.



2. Selanjutnya, klik *create* untuk melanjutkan konfigurasi *Virtual Network*.



3. Lanjutkan untuk konfigurasi IP di dalam *Virtual Network*.



4. Untuk *tab Basics* isi *Resource Group* dengan *Resource Group* yang baru, jika belum ada bisa *Create New*. Lalu untuk *Virtual network name* bisa diisi sesuai dengan kebutuhan. Untuk *Region* pilih *(Asia Pacific) Southeast Asia*. Lalu klik *next* untuk melanjutkan konfigurasi pada *tab Security*.

The screenshot shows the 'Create virtual network' page in the Microsoft Azure portal, specifically the 'Basics' tab. The page has a blue header with the Microsoft Azure logo and a search bar. Below the header, there's a breadcrumb trail: 'Home > Network foundation | Virtual networks'. The main title is 'Create virtual network'. Below this, there are tabs for 'Basics', 'Security', 'Address space', 'Tags', and 'Review + create'. The 'Basics' tab is active. Under 'Project details', there's a section to 'Select the subscription to manage deployed resources and code. Use resource groups like folders to organize and manage all your resources.' It shows 'Subscription' as 'Azure for Students' and 'Resource group' as '[New] MyLoadBalancer' with a 'Create new' link. Under 'Instance details', 'Virtual network name' is 'TA-Comp-vnet' and 'Region' is '(Asia Pacific) Southeast Asia' with a 'Deploy to an Azure Extended Zone' link. At the bottom, there are buttons for 'Previous', 'Next', and 'Review + create'.

5. Selanjutnya, pada *tab Security* bisa ceklis *Enable Azure Bastion*. Untuk *Azure Firewall*, bisa diceklis atau tidak. Selanjutnya klik *next* untuk konfigurasi *Address space*.

The screenshot shows the 'Create virtual network' page in the Microsoft Azure portal, specifically the 'Security' tab. The page has a blue header with the Microsoft Azure logo and a search bar. Below the header, there's a breadcrumb trail: 'Home > Network foundation | Virtual networks'. The main title is 'Create virtual network'. Below this, there are tabs for 'Basics', 'Security', 'Address space', 'Tags', and 'Review + create'. The 'Security' tab is active. A message says 'Enhance the security of your virtual network with these additional paid security services. Learn more'. Under 'Virtual network encryption', there's a description: 'Enable Virtual network encryption to encrypt traffic traveling within the virtual network. Virtual machines must have accelerated networking enabled. Traffic to public IP addresses is not encrypted. Learn more.' and a checkbox for 'Virtual network encryption' which is currently unchecked. Under 'Azure Bastion', there's a description: 'Azure Bastion is a paid service that provides secure RDP/SSH connectivity to your virtual machines over TLS. When you connect via Azure Bastion, your virtual machines do not need a public IP address. Learn more.' and a checkbox for 'Enable Azure Bastion' which is currently unchecked. Under 'Azure Firewall', there's a section header but no content is visible. At the bottom, there are buttons for 'Previous', 'Next', and 'Review + create'.

Create virtual network

Basics Security Address space Tags Review + create

Azure Bastion

Azure Bastion is a paid service that provides secure RDP/SSH connectivity to your virtual machines over TLS. When you connect via Azure Bastion, your virtual machines do not need a public IP address. [Learn more.](#)

Enable Azure Bastion



Azure Bastion host name

TA-demo-vnet-Bastion

Azure Bastion public IP address *

(New) ta-demo-vnet-bastion

[Create a public IP address](#)

Create virtual network

Basics Security Address space Tags Review + create

Enable Azure Firewall



Azure Firewall name

TA-demo-vnet-Firewall

Subnet name *

AzureFirewallSubnet

Tier *

Standard

Policy *

None (Use classic firewall rules)

[Create new](#)

Azure Firewall public IP address *

(New) ta-demo-vnet-firewall

[Create a public IP address](#)

Azure DDoS Network Protection

Azure DDoS Network Protection is a paid service that offers enhanced DDoS mitigation capabilities via adaptive tuning, attack notification, and telemetry to protect against the impacts of a DDoS attack for all protected resources within this virtual network. [Learn more.](#)

[Previous](#)[Next](#)[Review + create](#)

6. Selanjutnya untuk *tab Address Space* secara *default* akan menampilkan daftar Subnet yang dibuat.

The screenshot shows the 'Create virtual network' page in the Microsoft Azure portal, specifically the 'Address space' tab. The page title is 'Create virtual network'. Below the title, there are tabs for 'Basics', 'Security', 'Address space', 'Tags', and 'Review + create'. The 'Address space' tab is active. The main content area shows the address space configuration. At the top, there is a dropdown for the address space, currently set to '10.0.0.0/16'. Below this, there is a table showing the subnets. The table has columns for 'Subnets', 'IP address range', 'Size', and 'NAT gateway'. There are three subnets listed: 'default' (10.0.0.0 - 10.0.0.255, /24, 256 addresses), 'AzureBastionSubnet' (10.0.1.0 - 10.0.1.63, /26, 64 addresses), and 'AzureFirewallSubnet' (10.0.1.64 - 10.0.1.127, /26, 64 addresses). At the bottom, there is a button 'Add IPv4 address space'.

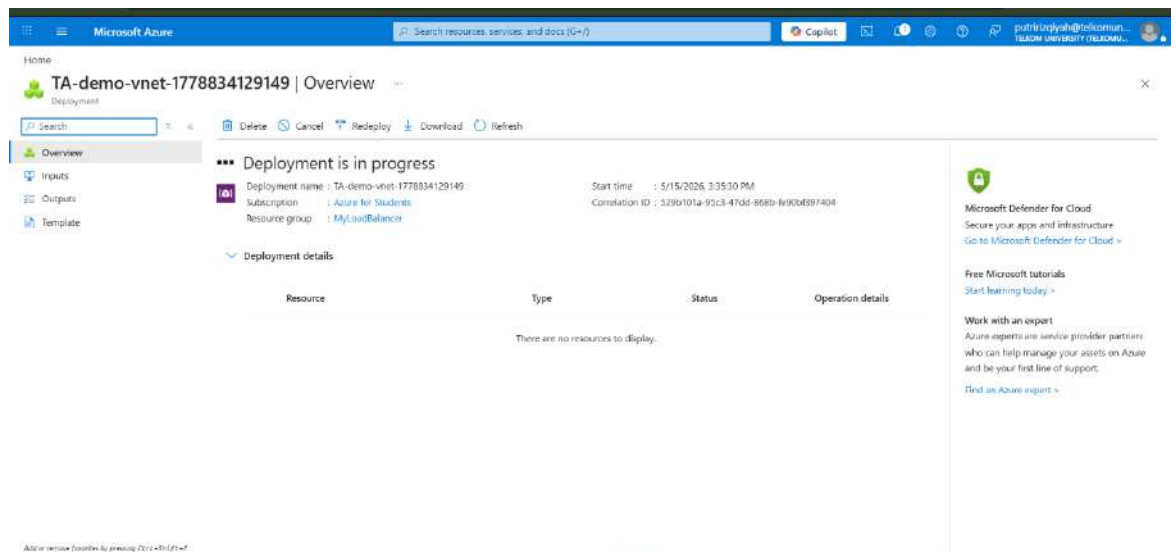
Subnets	IP address range	Size	NAT gateway
default	10.0.0.0 - 10.0.0.255	/24 (256 addresses)	-
AzureBastionSubnet	10.0.1.0 - 10.0.1.63	/26 (64 addresses)	-
AzureFirewallSubnet	10.0.1.64 - 10.0.1.127	/26 (64 addresses)	-

7. Selanjutnya klik next, untuk *Tab Tags* bisa dilewati dan bisa langsung menuju *Review+create*. Sebelum klik *create* bisa cek terlebih dahulu *summary* dari konfigurasi yang sudah dilakukan. Jika sudah yakin bisa klik *Create*.

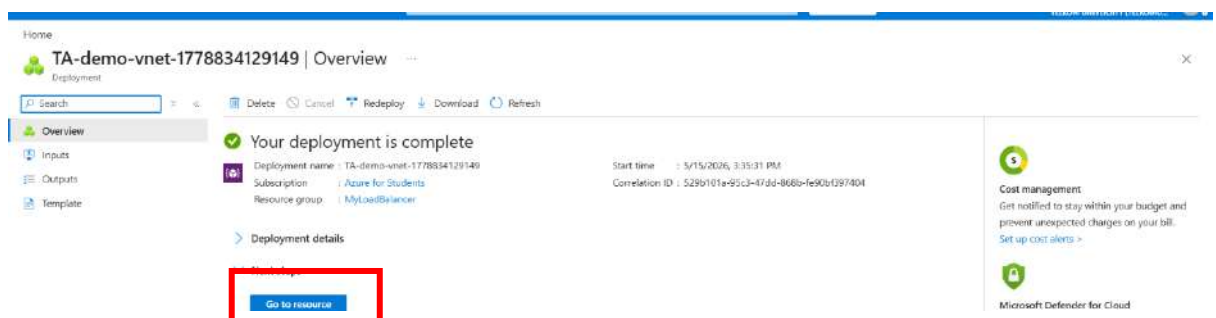
The screenshot shows the 'Create virtual network' page in the Microsoft Azure portal, specifically the 'Review + create' tab. The page title is 'Create virtual network'. Below the title, there are tabs for 'Basics', 'Security', 'Address space', 'Tags', and 'Review + create'. The 'Review + create' tab is active. The main content area shows a summary of the configuration. At the top, there is a green bar indicating 'Validation passed'. Below this, there is a table showing the configuration details. The table has columns for 'Category', 'Item', and 'Value'. The categories are 'Basics', 'Security', 'Address space', and 'Tags'. The items and values are as follows:

Category	Item	Value
Basics	Subscription	Azure for Students
	Resource Group	MyLoadBalancer
	Name	TA-demo-vnet
	Region	Southeast Asia
Security	Azure Bastion	Enabled
	- Name	(New) TA-demo-vnet-Bastion
	- Public IP Address	(New) ta-demo-vnet-bastion
	Azure Firewall	Enabled
Address space	- Name	(New) TA-demo-vnet-Firewall (Standard)
	- Public IP Address	(New) ta-demo-vnet-firewall
	Azure DDoS Network Protection	Disabled
	Address space	10.0.0.0/16 (65,536 addresses)
Tags	Subnet	default (10.0.0.0/24) (256 addresses)
	- Private subnet	Enabled
	Subnet	AzureBastionSubnet (10.0.1.0/26) (64 addresses)
	Subnet	AzureFirewallSubnet (10.0.1.04/26) (64 addresses)

8. Setelah klik *create* proses *deployment* akan berjalan.

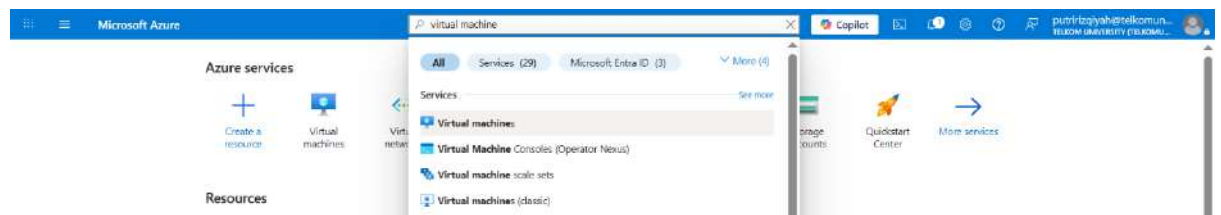


9. Setelah proses *Deployment* selesai klik *Go to resource* untuk melihat *overview* dari konfigurasi yang sudah dibuat. Lalu selanjutnya lakukan konfigurasi untuk membuat *Virtual Machine*.

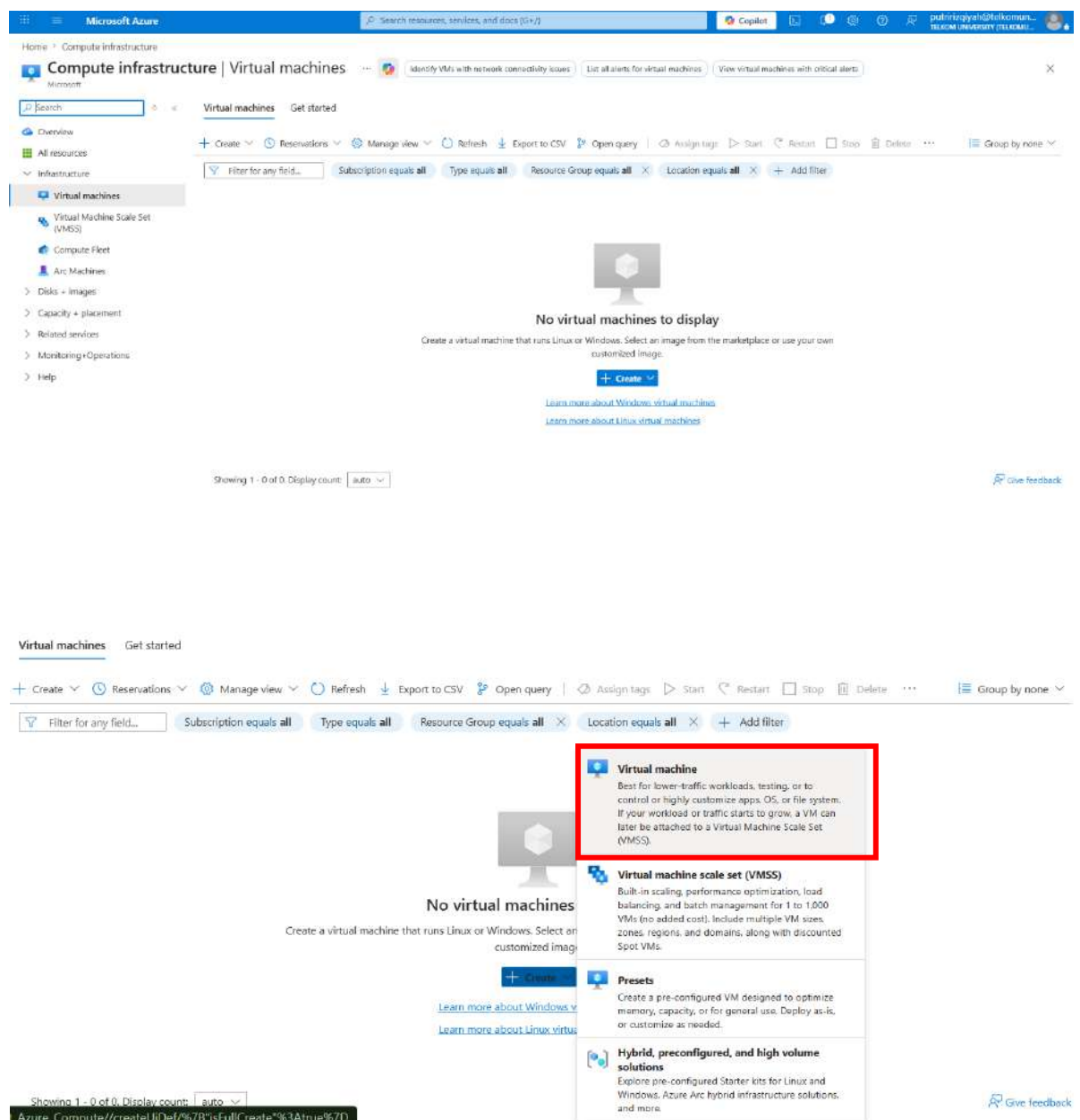


B. Konfigurasi Virtual Machine

1. Selanjutnya lakukan pencarian *Virtual Machine* pada *Search Bar*.



2. Selanjutnya klik *create* dan pilih *Virtual Machine*



3. Selanjutnya lakukan konfigurasi untuk *tab basics* seperti berikut ini. Beberapa hal yang harus diperhatikan adalah. *Region: (Asia Pacific) East Asia, Image: Windows Server 2022 Datacenter: Azure Edition Hotpatch -x64 Gen2, Size: Standard_B2ats_v2 – 2 cpus, 1 GiB memory (\$16.28) (free services eligible)*

The screenshot displays the 'Create a virtual machine' page in the Microsoft Azure portal, specifically the 'Basics' tab. The page is titled 'Create a virtual machine' and includes a search bar at the top. Below the title, there are tabs for 'Basics', 'Disks', 'Networking', 'Management', 'Monitoring', 'Advanced', 'Tags', and 'Review + create'. The 'Basics' tab is selected, and it contains a description: 'Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)'.

Below the description, there is a warning message: 'This subscription may not be eligible to deploy VMs of certain sizes in certain regions.' The 'Project details' section includes a dropdown for 'Subscription' (set to 'Azure for Students') and a dropdown for 'Resource group' (set to 'MyLoadBalancer'). The 'Instance details' section includes a dropdown for 'Virtual machine name' (set to 'TA-python-demo-vm'), a dropdown for 'Region' (set to '(Asia Pacific) East Asia'), a dropdown for 'Availability options' (set to 'Availability zone'), and a dropdown for 'Zone options' (set to 'Self-selected zone'). At the bottom of this section are buttons for '< Previous', 'Next > Disks', and 'Review + create'.

The second screenshot shows the 'Basics' tab configuration for the 'Create a virtual machine' page. It includes a dropdown for 'Availability zone' (set to 'Zone 1'), a dropdown for 'Security type' (set to 'Trusted launch virtual machines'), a dropdown for 'Image' (set to 'Windows Server 2022 Datacenter: Azure Edition Hotpatch - x64 Gen2'), and a dropdown for 'VM architecture' (set to 'x64'). Below these, there is a checkbox for 'Run with Azure Spot discount' (unchecked), a dropdown for 'Size' (set to 'Standard_B2ats_v2 - 2 vcpus, 1 GiB memory (\$16.28) (free services eligible)'), and a checkbox for 'Enable Hibernation' (unchecked). The 'Administrator account' section includes a dropdown for 'Username' (set to 'TA-demo1'), a dropdown for 'Password' (set to '*****'), and a dropdown for 'Confirm password' (set to '*****'). At the bottom of this section are buttons for '< Previous', 'Next > Disks', and 'Review + create'.

Compatible with Hyper-V to enable this feature. [Learn more](#)

Administrator account

Username *  ✓

Password * ✓

Confirm password * ✓

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports *  ☐ None
☒ Allow selected ports

Select inbound ports * ✓

⚠ This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

< Previous
Next : Disks >
Review + create

4. Selanjutnya lakukan konfigurasi untuk Disks, lalu klik *next*

Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me create a VM optimized for high availability
Help me create a low cost VM
Help me choose the right VM size for my workload

Basics
Disks
Networking
Management
Monitoring
Advanced
Tags
Review + create

Azure VMs have one operating system disk and a temporary disk for short-term storage. You can attach additional data disks. The size of the VM determines the type of storage you can use and the number of data disks allowed. [Learn more](#)

 There is a charge for the underlying storage resources consumed by your virtual machine. [Learn more](#)

VM disk encryption

Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host  ☐

 Encryption at host is not registered for the selected subscription. [Learn more](#)

OS disk

OS disk size 

OS disk type * 

Delete with VM  ☒

Key management 

Enable Ultra Disk compatibility  ☐

Data disks for TA-python-demo-vm

You can add and configure additional data disks for your virtual machine or attach existing disks. This VM also comes with a temporary disk.

LUN	Name	Size (GiB)	Disk type	Host caching	Delete with VM 
Create and attach a new disk Attach an existing disk					

< Previous
Next : Networking >
Review + create

5. Selanjutnya lakukan konfigurasi *tab Networking*, pada *tab* ini sebenarnya semuanya sudah ter-set secara default, yang harus ditambahkan adalah *Select inbound ports*, tambahkan HTTP (80) sebagai *inbound ports*, lalu ceklis *Delete public IP and NIC when VM is deleted*. Jika sudah sesuai bisa klik next

The screenshot shows the 'Networking' tab in the 'Create a virtual machine' wizard. The 'Virtual network' is set to '(new) TA-python-demo-vm-vnet', 'Subnet' to '(new) default (10.0.0.0/24)', and 'Public IP' to '(new) TA-python-demo-vm-ip'. Under 'NIC network security group', 'Basic' is selected. Under 'Public inbound ports', 'Allow selected ports' is selected, and the 'Select inbound ports' dropdown shows 'HTTP (80), RDP (3389)'. A warning message states: 'This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.' The checkbox 'Delete public IP and NIC when VM is deleted' is checked. Navigation buttons at the bottom include '< Previous', 'Next: Management >', and 'Review + create'.

6. Selanjutnya lakukan konfigurasi untuk *tab Management*, pada *tab* ini sebenarnya tidak dilakukan konfigurasi apapun sehingga konfigurasi *tab Management*, *Monitoring*, *Advanced*, *Tags* bisa dilewati

The screenshot shows the 'Management' tab in the 'Create a virtual machine' wizard. It includes sections for 'Microsoft Defender for Cloud' (with 'Enable basic plan for free' checked), 'Identity' (with 'Enable system assigned managed identity' unchecked), 'Microsoft Entra ID' (with 'Login with Microsoft Entra ID' unchecked and a note about RBAC), 'Auto-shutdown' (unchecked), 'Backup' (unchecked), and 'Site Recovery' (unchecked). Navigation buttons at the bottom include '< Previous', 'Next: Monitoring >', and 'Review + create'.

Microsoft Azure

Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me create a VM optimized for high availability Help me create a k

Basics Disks Networking Management **Monitoring** Advanced Tags Review + create

Configure monitoring options for your VM.

Alerts

Enable recommended alert rules ☐

Diagnostics

Boot diagnostics ☒ Enable with managed storage account (recommended)
☐ Enable with custom storage account
☐ Disable
 There is a nominal charge for the managed storage account.

Enable OS guest diagnostics ☐

Health

Enable application health monitoring ☐

< Previous Next: Advanced > Review + create

7. Berikut *preview* dari summary konfigurasi yang sudah dilakukan, selanjutnya klik *create*

Microsoft Azure

Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me create a VM optimized for high availability Help me create a low cost VM Help me ch

Validation passed

Basics Disks Networking Management Monitoring Advanced Tags **Review + create**

Price

1 X Standard B2ats v2
 by Microsoft
[Terms of use](#) | [Privacy policy](#)

Subscription credits apply ☐
0.0223 USD/hr
[Pricing for other VM sizes](#)

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

You have set RDP port(s) open to the internet. This is only recommended for testing. If you want to change this setting, go back to Basics tab.

Basics

Subscription	Azure for Students
Resource group	MyLoadBalancer
Virtual machine name	TA-python-demo-vm
Region	East Asia
Availability options	Availability zone

< Previous Next > Create

Microsoft Azure

Search

Home

>

Compute infrastructure

>

Virtual machines

Create a virtual machine

Help me create a VM optimized for high availability

Help me create a

Validation passed

Availability options	Availability zone
Zone options	Self-selected zone
Availability zone	1
Security type	Trusted launch virtual machines
Enable secure boot	Yes
Enable vTPM	Yes
Integrity monitoring	No
Image	Windows Server 2022 Datacenter Azure Edition Hotpatch - Gen2
VM architecture	x64
Size	Standard B2ats v2 (2 vcpus, 1 GiB memory)
Enable Hibernation	No
Username	TA-demo1
Public inbound ports	RDP, HTTP
Already have a Windows license?	No
Azure Spot	No

Disks

OS disk size	Image default
OS disk type	Premium SSD LRS
Use managed disks	Yes
Delete OS disk with VM	Enabled
Ephemeral OS disk	No

Networking

Virtual network	(new) TA-python-demo-vm-vnet
-----------------	------------------------------

< Previous

Next >

Create

Microsoft Azure

Search

Home

>

Compute infrastructure

>

Virtual machines

Create a virtual machine

Help me create a VM optimized for high availat

Validation passed

Networking

Virtual network	(new) TA-python-demo-vm-vnet
Subnet	(new) default (10.0.0.0/24)
Public IP	(new) TA-python-demo-vm-ip
Accelerated networking	On
Place this virtual machine behind an existing load balancing solution?	No
Delete public IP and NIC when VM is deleted	Enabled

Management

Microsoft Defender for Cloud	Basic (free)
System assigned managed identity	Off
Login with Microsoft Entra ID	Off
Auto-shutdown	Off
Backup	Disabled
Site Recovery	Disabled
Enable periodic assessment	Off
Enable hotpatch	On
Patch orchestration options	Azure-orchestrated patching (preview): patches will be install
Reboot setting	Reboot if required

Monitoring

Alerts	Off
--------	-----

< Previous

Next >

Create

Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me create a VM optimized for high availability | Help me

Validation passed

Microsoft Entra ID	None
System assigned managed identity	Off
Login with Microsoft Entra ID	Off
Auto-shutdown	Off
Backup	Disabled
Site Recovery	Disabled
Enable periodic assessment	Off
Enable hotpatch	On
Patch orchestration options	Azure-orchestrated patching (preview): patches will be installed by Azure
Reboot setting	Reboot if required

Monitoring

Alerts	Off
Boot diagnostics	On
Enable OS guest diagnostics	Off
Enable application health monitoring	Off

Advanced

Extensions	None
VM applications	None
Cloud init	No
User data	No
Disk controller type	SCSI
Proximity placement group	None
Capacity reservation group	None

< Previous Next > **Create**

8. Berikut proses *deployment* dari *Virtual Machine*.

Microsoft Azure

Home

CreateVm-microsoftwindowsserver.windowsserver2022-20260515160703 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

*** Deployment is in progress

Deployment name: CreateVm-microsoftwindowsserver.windowsserver2022-20260515160703

Subscription: Azure for Students

Resource group: MyCodeBalancer

Start time: 5/15/2026, 4:17:29 PM

Correlation ID: 36930817-8607-40e2-8ab3-9906d7a3dd1e

Deployment details

Resource	Type	Status	Operation details
ts-python-demo-vnet	Virtual network	OK	Operation details
ts-python-demo-nsg	Network security group	OK	Operation details
ts-python-demo-vm	Public IP address	OK	Operation details

Next steps

9. Proses *deployment selesai*, selanjutnya klik *Go To Resource*

Microsoft Azure

Home

CreateVm-microsoftwindowsserver.windowsserver2022-20260515160703 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name: CreateVm-microsoftwindowsserver.windowsserver2022-20260515160703

Subscription: Azure for Students

Resource group: MyCodeBalancer

Start time: 5/15/2026, 4:18:05 PM

Correlation ID: 36930817-8607-40e2-8ab3-9906d7a3dd1e

Deployment details

Next steps

Set up auto shutdown Recommended

Monitor VM health, performance, and network dependencies Recommended

Run a script inside the virtual machine Recommended

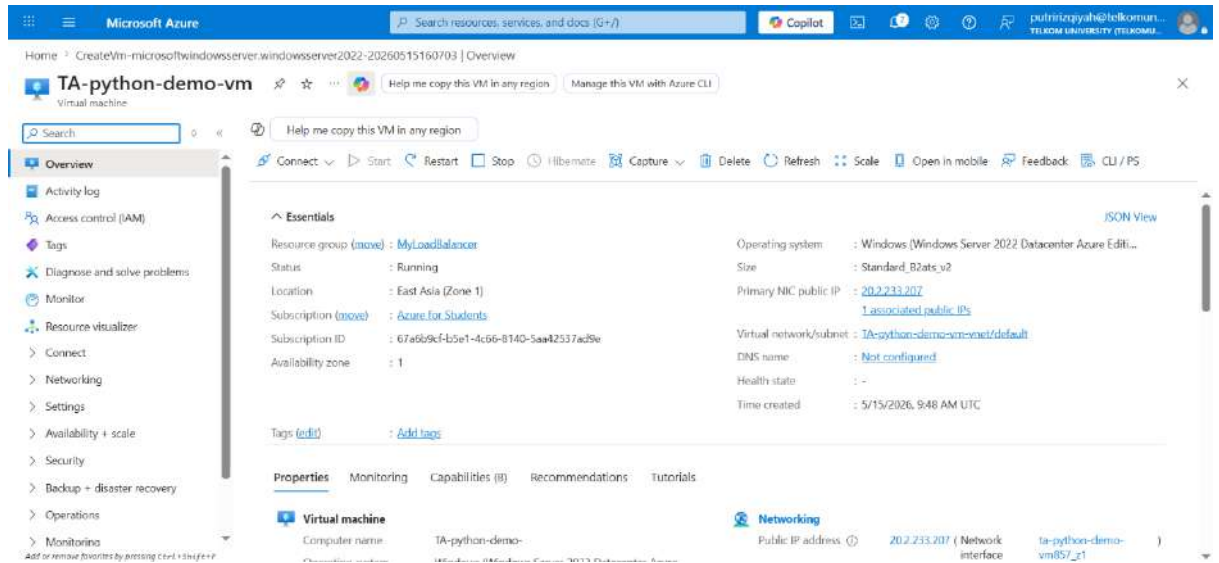
Go to resource Create another VM Scale out your VM

Cost management

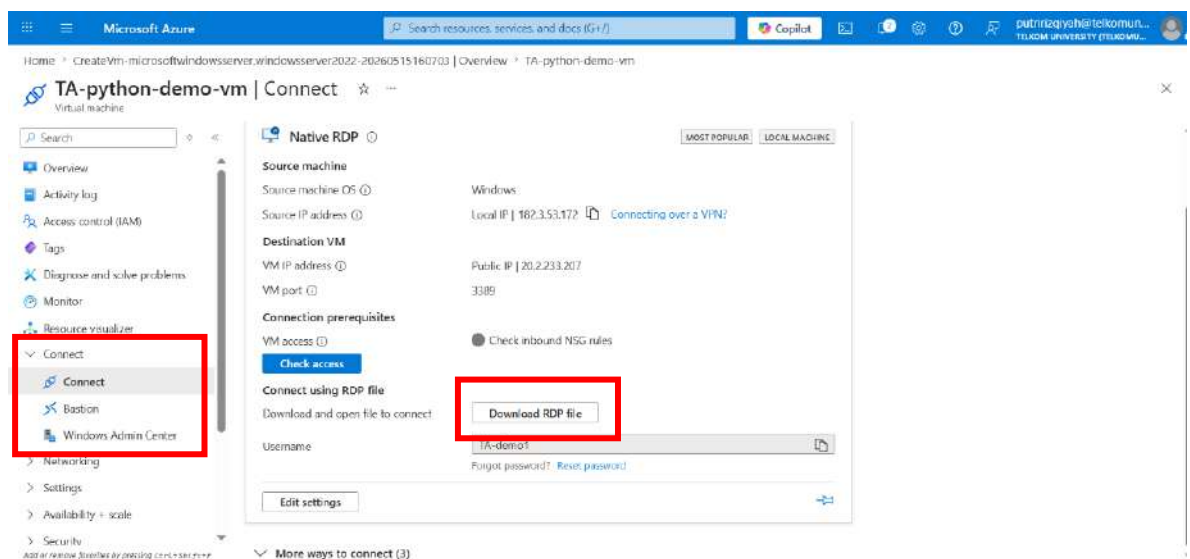
Get notified to stay within your budget and prevent unexpected charges on your bill.

Set up cost alerts

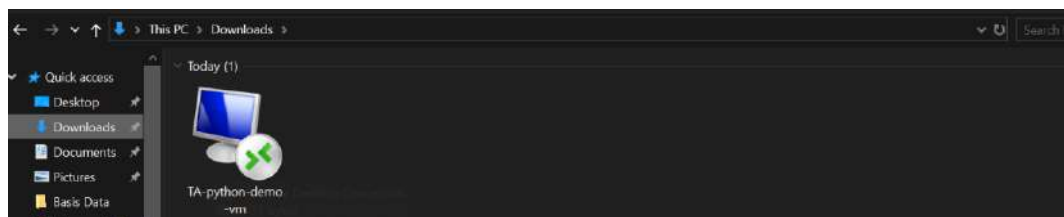
10. Berikut overview dari *Virtual Machine* yang sudah dibuat



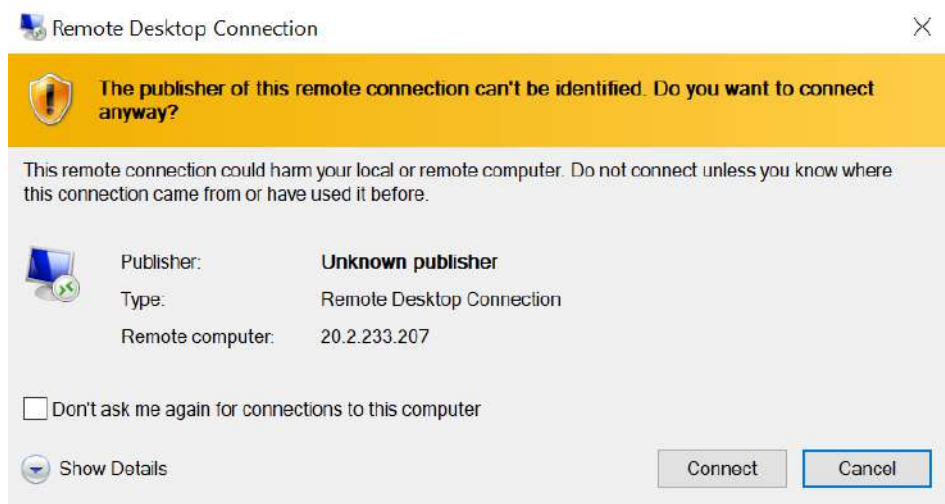
11. Selanjutnya klik menu *connect* dan klik *download RDP file*.



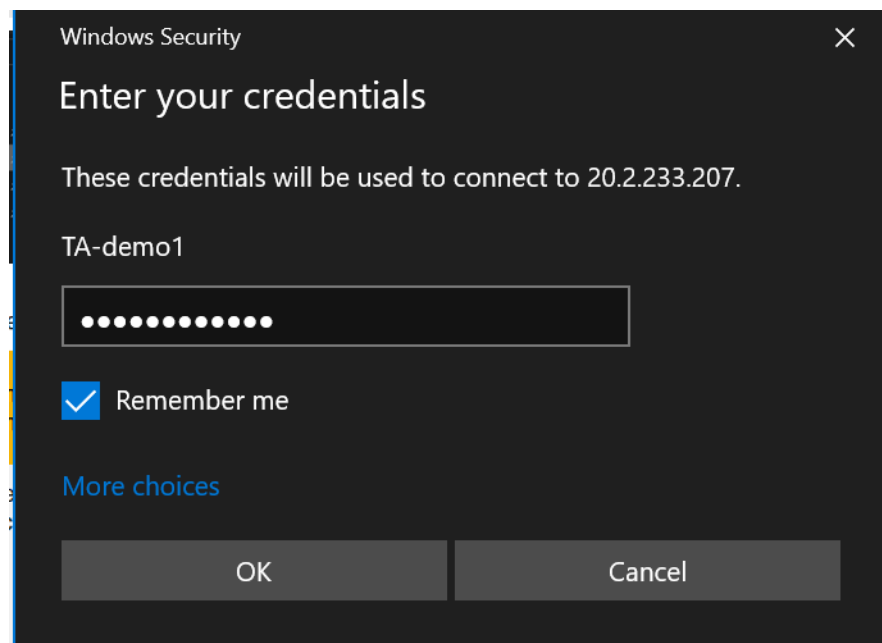
12. Berikut tampilan dari *RDP file* yang akan digunakan untuk terhubung dengan *Windows Server* menggunakan *Remote Desktop*



13. Double klik pada file RDP yang sudah di-*download* tadi dan klik *connect*



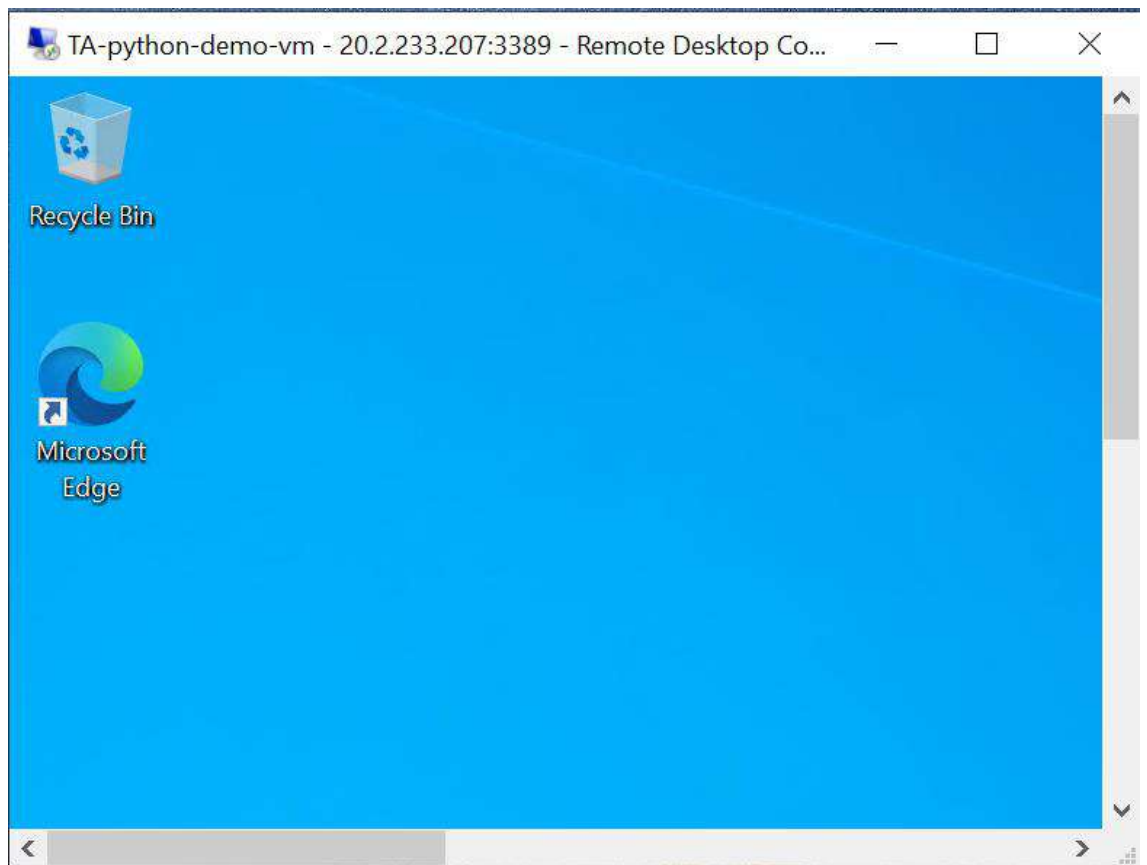
14. Masukkan *password* yang digunakan saat konfigurasi *Virtual Machine* pada *tab basics* di Azure



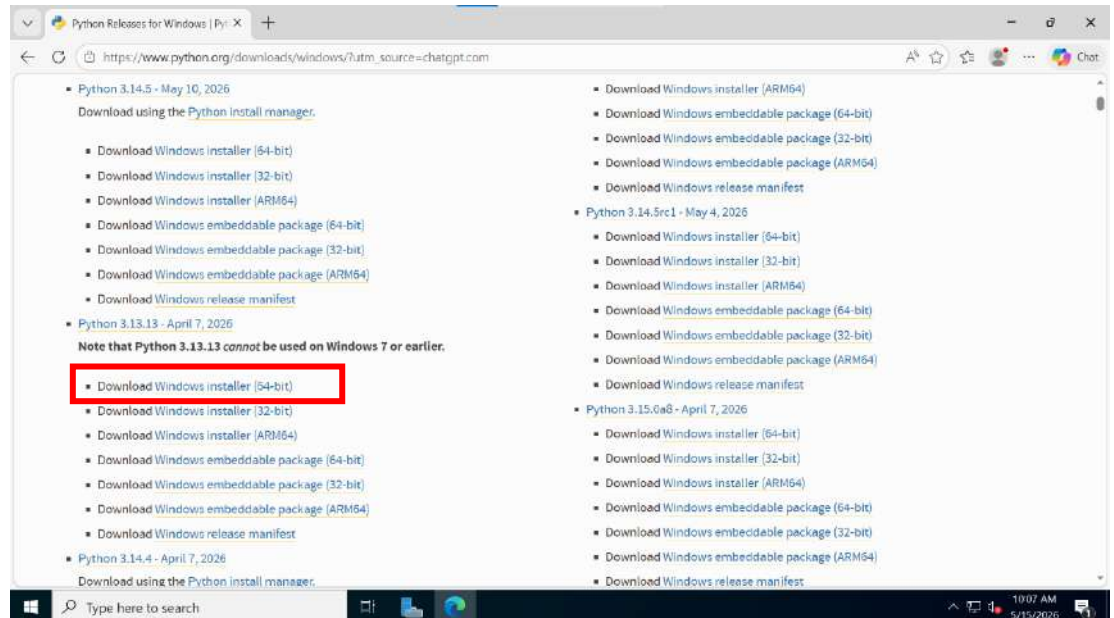
15. Selanjutnya klik Yes



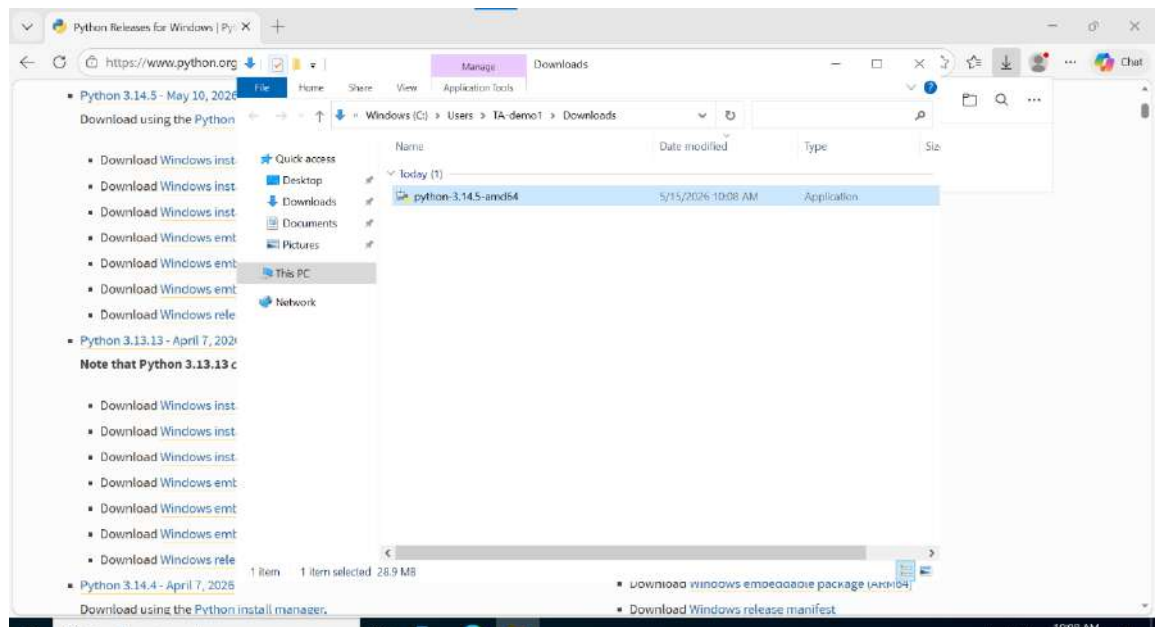
16. Berikut tampilan ketika sukses masuk ke dalam Windows Server



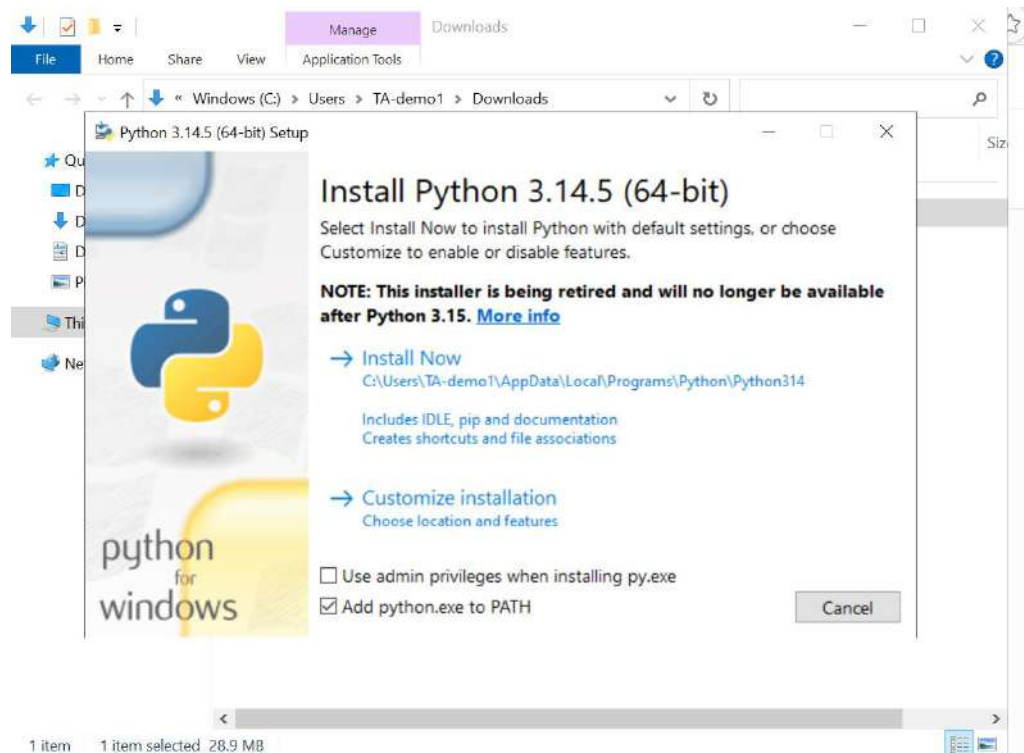
17. Sekarang, lakukan instalasi Python di dalam *Windows Server*. Buka Microsoft Edge Browser dan klik link berikut ini:
http://python.org/downloads/windows/?utm_source=chatgpt.com
Pilih python *Windows Installer (64-bit)*.



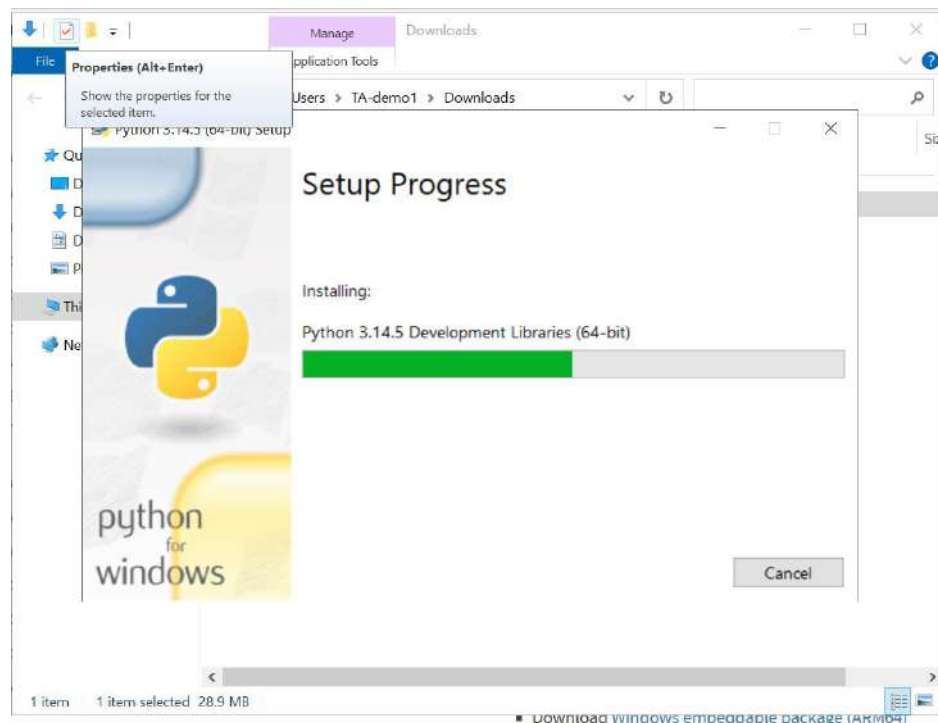
18. Selanjutnya, setelah file *installer* berhasil ter-*download* double klik dan lakukan instalasi seperti pada umumnya.



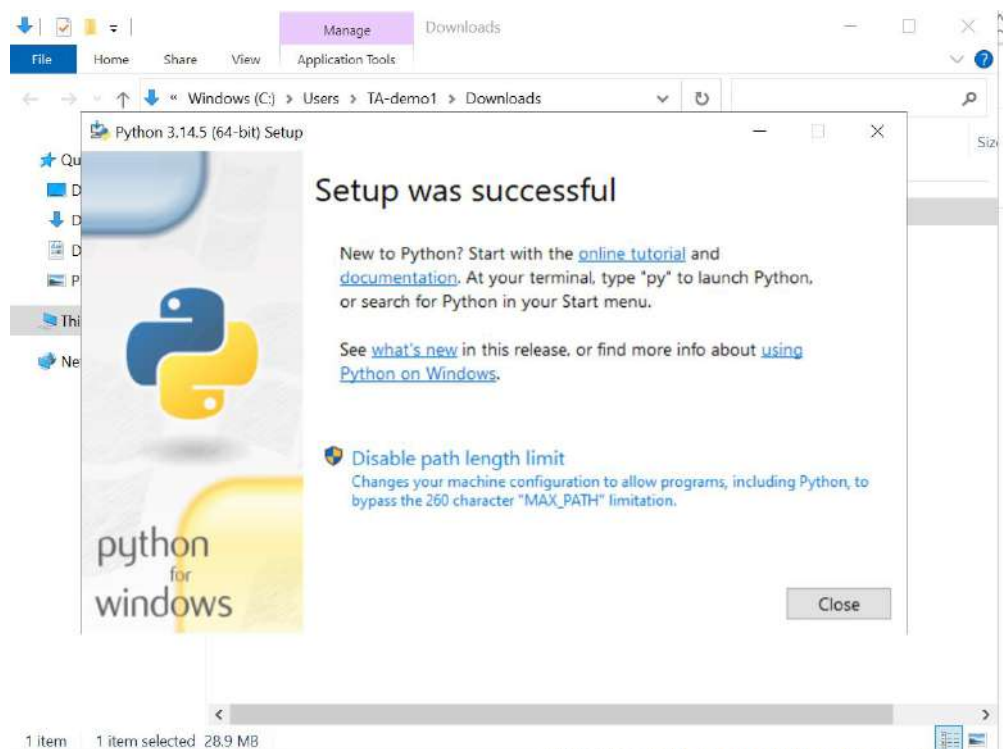
19. Sebelum melanjutkan proses instalasi jangan lupa untuk **CEKLIS Add python.exe to PATH** supaya *python* bisa dijalankan lewat *Command Line* (CMD)



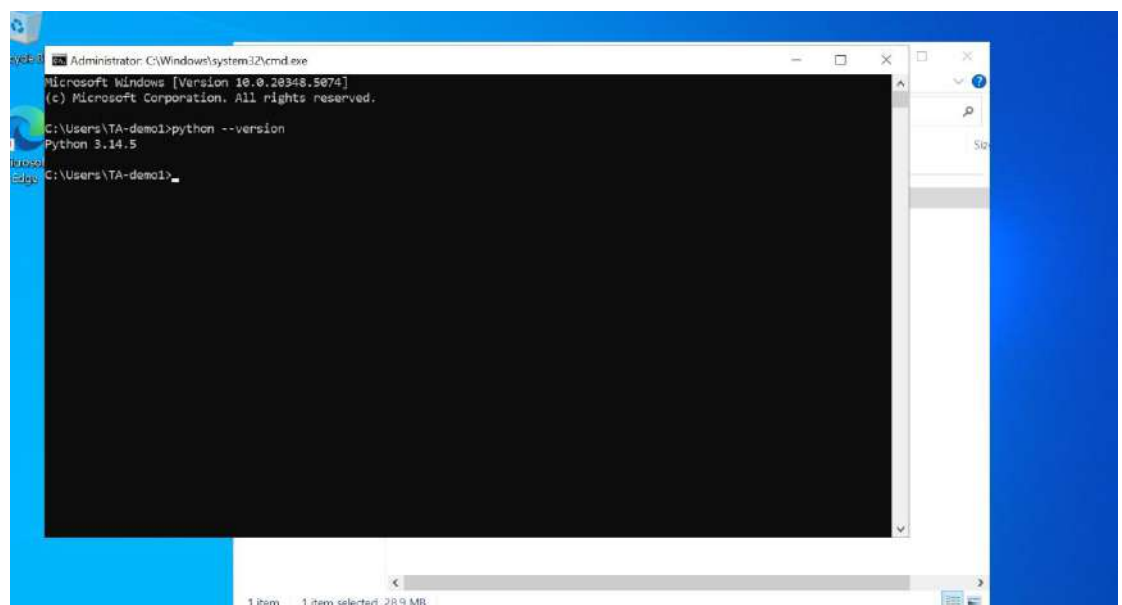
20. Berikut proses instalasi python di dalam Windows Server



21. Jika sudah muncul tampilan berikut, python sudah berhasil terinstall



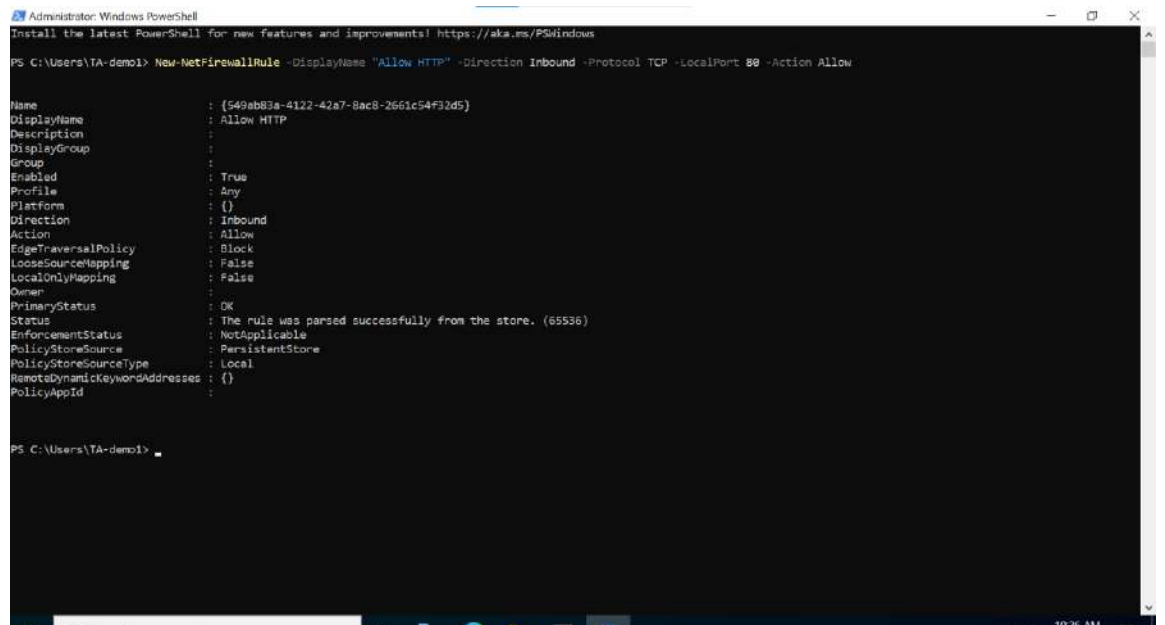
22. Untuk memastikan python berhasil terinstall buka Command Line (*windows+r*) ketike cmd dan lakukan pengecekan versi python dengan menulis perintah *python --version*. Jika sudah muncul versi python, maka python sudah sukses terinstall.



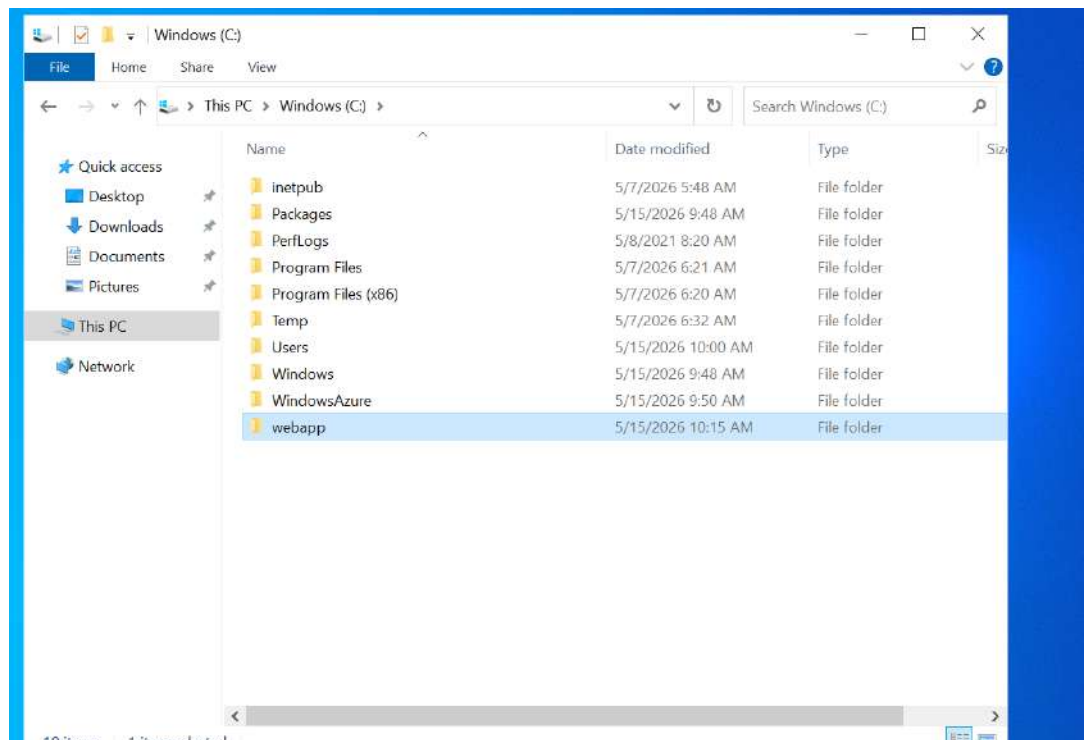
23. Selanjutnya buka windows Powershell



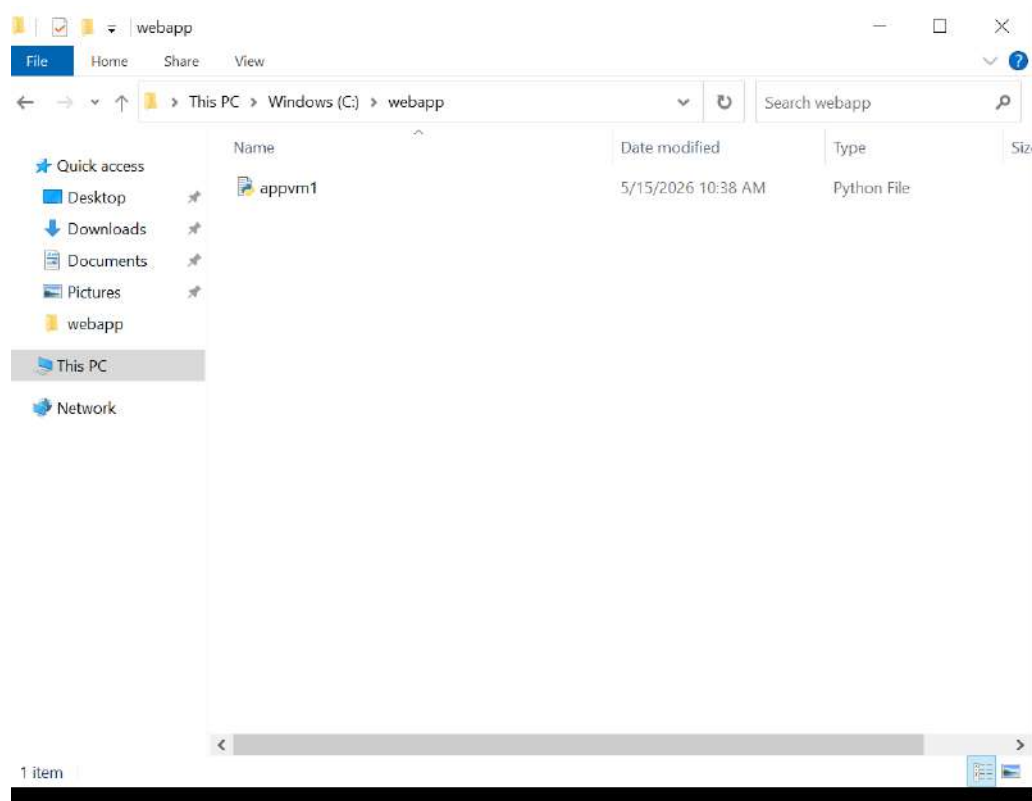
24. Lalu masukkan perintah berikut pada PowerShell untuk mengaktifkan Port 80 saat akses backend server `New-NetFirewallRule -DisplayName "Allow HTTP" -Direction Inbound -Protocol TCP -LocalPort 80 -Action Allow`



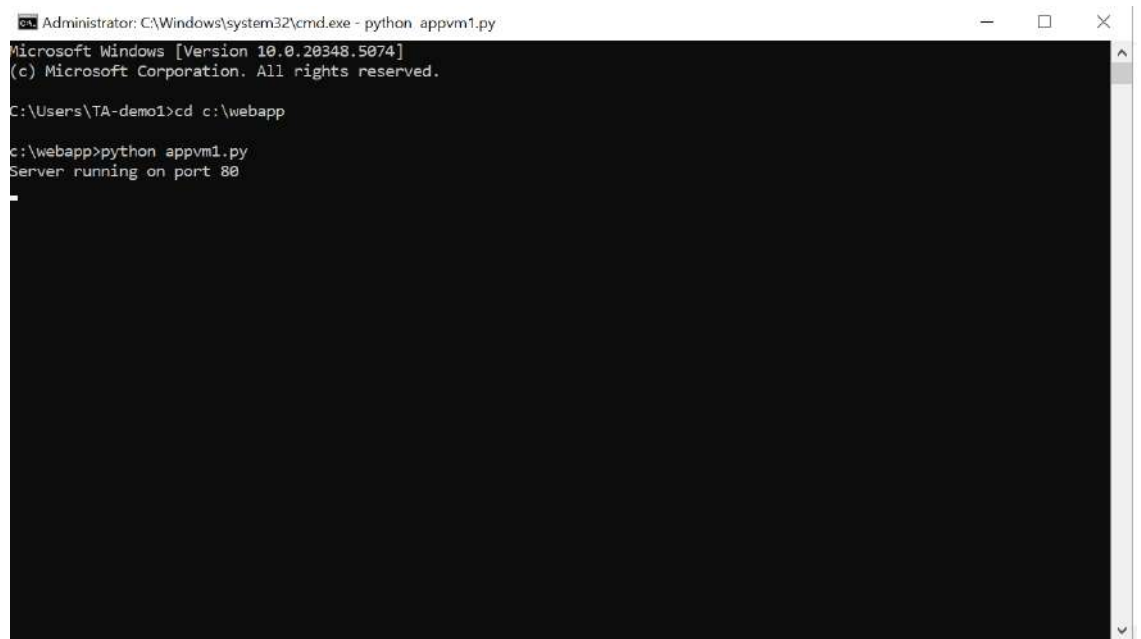
25. Selanjutnya masuk ke disk C: di dalam *file explorer* dan buat *folder webapp*



26. Lakukan Copy-pasti file `app2.py` dari lokal laptop kita ke dalam windows server (file sudah tersedia di lms, bisa langsung download dan copy paste ke dalam disk C: Windows Server).



27. Selanjutnya jalankan file python dengan *script* berikut *python appvm1.py* -> nama file dari pyhton project.

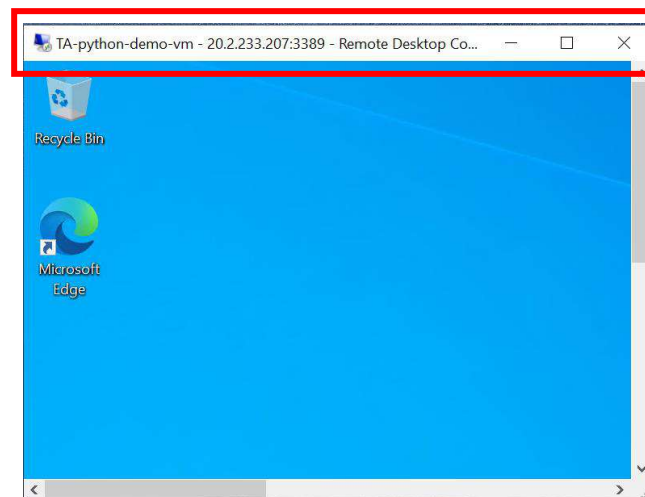


```
Administrator: C:\Windows\system32\cmd.exe - python appvm1.py
Microsoft Windows [Version 10.0.20348.5074]
(c) Microsoft Corporation. All rights reserved.

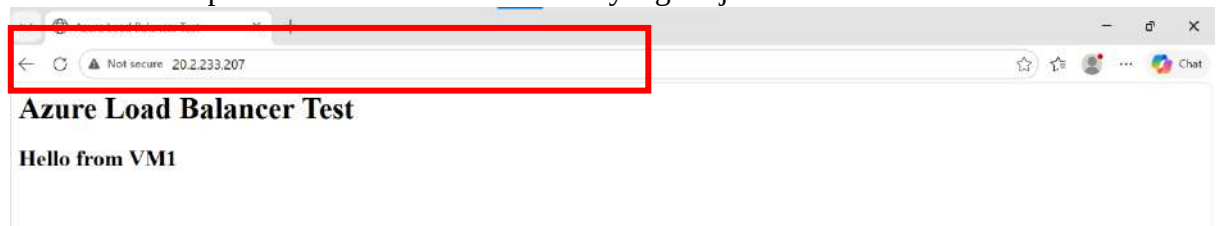
C:\Users\TA-demo1>cd c:\webapp

c:\webapp>python appvm1.py
Server running on port 80
```

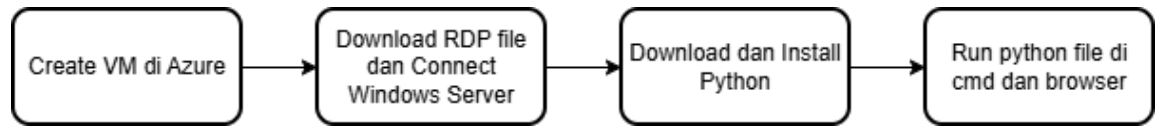
28. Selanjutnya buka browser dan masukkan public ip yang didapatkan dari Azure. Public IP dapat dilihat dengan cara masuk ke Overview Virtual Machine di dalam Azure, lalu copy paste public ip ke dalam browser, atau sebagai opsi lain public IP akan terlihat saat connect Remote Dekstop.



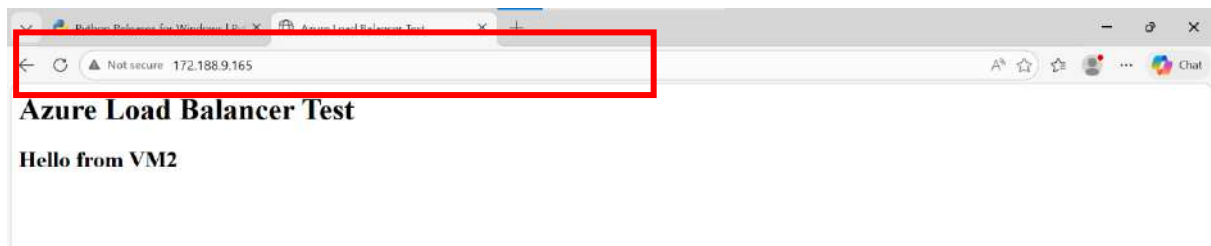
29. Berikut tampilan dari web statis sederhana yang berjalan di VM 1



30. Selanjutnya, lakukan hal yang sama untuk membuat Virtual Machine yang kedua. Mulai dari proses konfigurasi di dalam Azure sampai proses akses file python melalui command line dan browser.

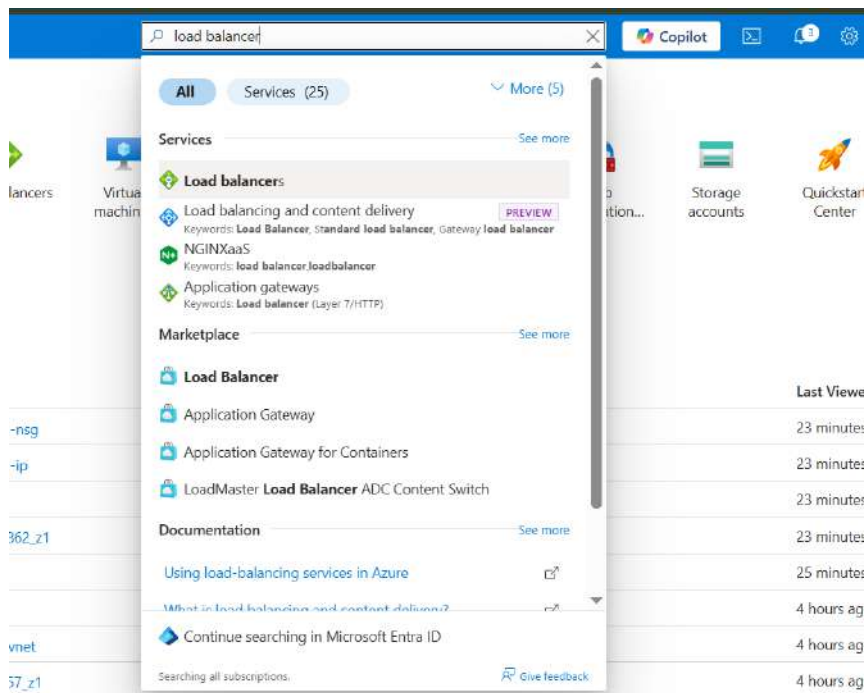


Berikut tampilan python file dari Virtual Machine yang kedua. Bisa dilihat keduanya (VM1 dan VM2) memiliki IP yang berbeda menandakan bahwasannya kedua file tersebut diakses menggunakan IP yang berbeda

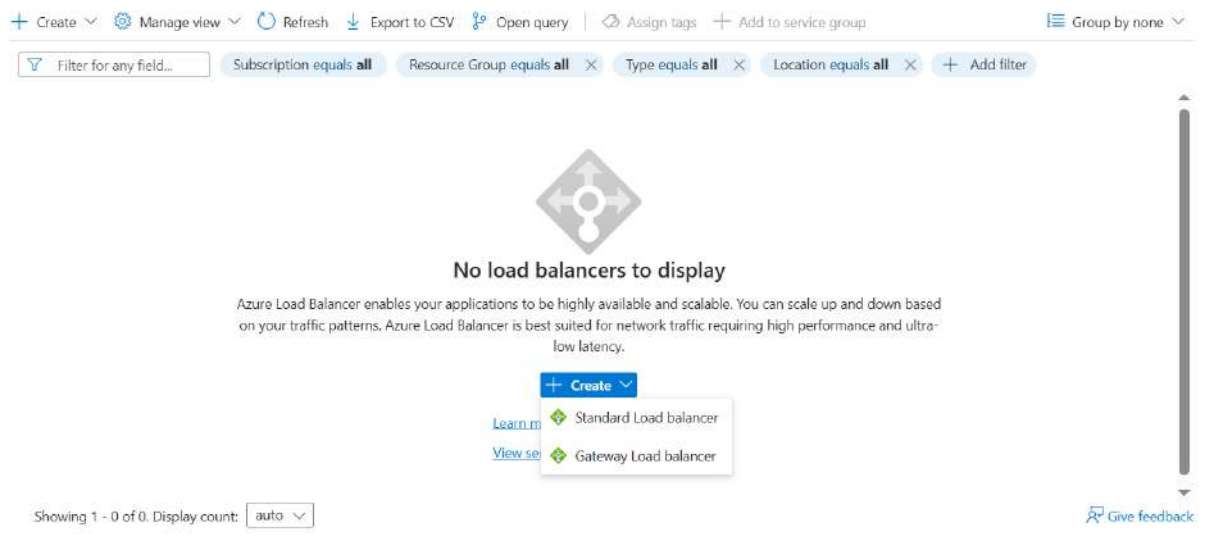


C. Membuat Load Balancer

1. Selanjutnya adalah membuat Load Balancer, ketik Load Balancer pada *search bar* di dalam *Azure*.



2. Lalu create Standard Load Balancer



3. Lakukan konfigurasi untuk masing-masing tab seperti berikut ini

Home > Load balancing and content delivery | Load balancers

Create load balancer

Basics Frontend IP configuration Backend pools Inbound rules Outbound rules Tags Review + create

Azure load balancer is a layer 4 load balancer that distributes incoming traffic among healthy virtual machine instances. Load balancers uses a hash-based distribution algorithm. By default, it uses a 5-tuple (source IP, source port, destination IP, destination port, protocol type) hash to map traffic to available servers. Load balancers can either be internet-facing where it is accessible via public IP addresses, or internal where it is only accessible from a virtual network. Azure load balancers also support Network Address Translation (NAT) to route traffic between public and private IP addresses. [Learn more.](#)

Project details

Subscription * Azure for Students

Resource group * MyLoadBalancer

[Create new](#)

Instance details

Name * TA-demo-loadbalancer

Region * East Asia

SKU * ☒ Standard (Distribute traffic to backend resources)
☐ Gateway (Direct traffic to network virtual appliances)

Type * ☒ Public
☐ Internal

Tier * ☒ Regional
☐ Global

[Review + create](#) [< Previous](#) [Next : Frontend IP configuration >](#) [Download a template for automation](#) [Give feedback](#)

4. Selanjutnya, lakukan konfigurasi Frontend IP Configuration seperti berikut ini. Klik **Add a frontend IP Configuration**. Lalu isi field yang perlu diisi seperti berikut ini.

Home > Load balancing and content delivery | Load balancers

Create load balancer

Basics **Frontend IP configuration** Backend pools Inbound rules Outbound rules Tags Review & create

A frontend IP configuration is an IP address used for inbound and/or outbound communication as defined within load balancing, inbound NAT, and outbound rules.

+ Add a frontend IP configuration

Name: frontend-ip IP address: frontend-ip

Add a frontend IP to get started

Review & create < Previous Next: Backend pools > Download a template for automation / Give feedback

Microsoft Azure Search resources, services and docs (v2) Copilot Publishing with Microsoft Azure DevOps (v2)

Add frontend IP configuration

Name * frontend-ip

IP version ☒ IPv4 ☐ IPv6

IP type ☒ IP address ☐ IP prefix

Public IP address * Select public IP address
 [Create new](#)

Gateway Load balancer ☐ None

Save Cancel Give feedback

Add a public IP address

Name * TA-demo-lb-ip

SKU Standard

Tier Regional

Static IPs are assigned at the time the resource is created and released when the resource is deleted.

Assignment ☒ Static ☐ Dynamic

Availability zone * Zone-redundant

Routing preference ☒ Microsoft network ☐ Internet

Save Cancel

Microsoft Azure

Search resources, services, and docs (G+/I)

Home > Load balancing and content delivery | Load balancers

Create load balancer

Basics **Frontend IP configuration** Backend pools Inbound rules Outbound rules Tags Review + create

A frontend IP configuration is an IP address used for inbound and/or outbound communication as defined within load balancing, inbound NAT, and outbound rules.

+ Add a frontend IP configuration

Name ↑↓	IP address ↑↓
frontend-ip	(new) TA-demo-lb-ip (To be created)

Review + create < Previous Next : Backend pools > Download a template for automation Give feedback

- Selanjutnya, lakukan konfigurasi untuk tab *backend pool*, pada tahapan ini, masukkan Virtual Machine 1 dan Virtual Machine 2. Klik Add dan ceklis kedua VM yang sudah dibuat. **CATATAN: PASTIKAN VM1 DAN VM2 BERADA DI VIRTUAL NETWORK YANG SAMA.** Jika pada saat klik add hanya ada satu VM hal tersebut menandakan bahwasannya kedua VM yang dibuat tidak berada di Virtual Network yang sama!

Microsoft Azure

Search resources, services, and docs (G+/I)

Home > Create load balancer

Add backend pool

Name * TA-lbp-loadbalancer

Virtual network TA-python-demo-vm-vnet (MyLoadBalancer)

Backend Pool Configuration

☒ NIC ☐ IP address

IP configurations

IP configurations associated to virtual machines and virtual machine scale sets must be in same location as the load balancer and be in the same virtual network.

+ Add | X Remove

Resource Name	Resource group	Type	IP configuration	IP Address	Availability set
---------------	----------------	------	------------------	------------	------------------

Save Cancel Give feedback

Microsoft Azure

Search resources, services, and docs (Ctrl)

Copilot

publirzgyah@fakomun...
FUDOM UNIVERSITY (fudomun...)

Home > Create load balancer

Add backend pool

Name *

TA-bp-loadbalancer

Virtual network

TA-python-demo-vm-vnet (MyLoa...)

The dropdown only shows virtu...

Backend Pool Configuration

☒ NIC
 ☐ IP address

IP configurations

IP configurations associated to virtual machines and virtual machine scale sets must be in same location as the load balancer and be in the same virtual network.

+ Add

- Remove

Resource Name

Resource group

Add IP configurations to backend pool

Filter by name...

Location: eastasia

Virtual network: TA-python-demo-vm-vnet

Add filter

☐ Show resources that are not available for selection

Resource Name	Resource group	Type	IP configurati...	IP Address	Availability set	Tags
Virtual machine (2)						
☒ TA-PYTHON-DEMO-...	MYLOADBALA...	Virtual machine	ipconfig1	10.0.0.4	-	-
☒ TA-python-demo-v...	MyloadBalanc...	Virtual machine	ipconfig1	10.0.0.5	-	-

Add

Cancel

Save

Cancel

Give feedback

Microsoft Azure

Search resources, services, and docs (Ctrl)

Copilot

publirzgyah@fakomun...
FUDOM UNIVERSITY (fudomun...)

Home

Create load balancer

Basics

Frontend IP configuration

Backend pools

Inbound rules

Outbound rules

Tags

Review + create

A backend pool is a collection of resources to which your load balancer can send traffic. A backend pool can contain virtual machines, virtual machine scale sets, and containers.

+ Add a backend pool

Name	Virtual network	Resource Name	Network interface	IP address	Availability zone	Admin state
TA-bp-loadbalancer						
TA-bp-loadbalancer	TA-python-demo-vm-vnet	TA-PYTHON-DEMO-VM	ta-python-demo-vm\$7_21	10.0.0.4	1	None
TA-bp-loadbalancer	TA-python-demo-vm-vnet	TA-python-demo-vm-2rev	ta-python-demo-vm-2rev\$300_21	10.0.0.5	1	None

Review + create

< Previous

Next: Inbound rules >

Download a template for automation

Give feedback

6. Selanjutnya konfigurasi Inbound Rules. Pada tab ini kita melakukan konfigurasi Inbound Rules dan juga konfigurasi Health Probe, ikuti tahapan di bawah ini untuk konfigurasi Inbound Rules. Klik Add a load balancing rule lalu isi field yang kosong seperti berikut.

Microsoft Azure

Home

Create load balancer

Basics Frontend IP configuration Backend pools Inbound rules Outbound rules Tags Review + create

Load balancing rule

A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. The load balancing rule uses a health probe to determine which backend instances are eligible to receive traffic.

+ Add a load balancing rule

Name * Frontend IP configuration * Backend pool * Health probe *

Add a rule to get started

Inbound NAT rule

An inbound NAT rule forwards incoming traffic sent to a selected IP address and port combination to a specific virtual machine.

+ Add an inbound nat rule

Name * Frontend IP configuration * Service * Target *

Add a rule to get started

Review + create < Previous Next: Outbound rule > Download a template for automation Give feedback

Add load balancing rule

A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. The load balancing rule uses a health probe to determine which backend instances are eligible to receive traffic.

Name * load balancing rule name

IP version * ☒ IPv4 ☐ IPv6

Frontend IP address * Select an existing frontend IP address

Backend pool * Select an existing backend pool

Protocol ☒ TCP ☐ UDP

Port *

Backend port *

Health probe * No existing probes

Save Cancel Give feedback

Microsoft Azure

Home

Create load balancer

Basics Frontend IP configuration Backend pools Inbound rules Outbound rules Tags Review + create

Load balancing rule

A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. The load balancing rule uses a health probe to determine which backend instances are eligible to receive traffic.

+ Add a load balancing rule

Name * TA-ib-rule

IP version * ☒ IPv4 ☐ IPv6

Frontend IP address * frontend-ip (to be created)

Backend pool * TA-ib-loadbalancer

Protocol ☒ TCP ☐ UDP

Port * 80

Backend port * 80

Health probe * No existing probes

Save Cancel Give feedback

Microsoft Azure

Home

Create load balancer

Basics Frontend IP configuration Backend pools Inbound rules Outbound rules Tags Review + create

Load balancing rule

A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. The load balancing rule uses a health probe to determine which backend instances are eligible to receive traffic.

+ Add a load balancing rule

Name	Frontend IP configuration	Backend pool	Health probe	Frontend Port	Backend Port
TA-ib-rule	frontend-ip	TA-ib-loadbalancer	TA-ib-lb	80	80

Inbound NAT rule

An inbound NAT rule forwards incoming traffic sent to a selected IP address and port combination to a specific virtual machine.

+ Add an inbound nat rule

Name	Frontend IP configuration	Service	Target	Frontend Port
------	---------------------------	---------	--------	---------------

Review + create < Previous Next: Outbound rule > Download a template for automation Give feedback

7. Selanjutnya, untuk tab Outbound Rules dan Tags bisa dilewati dan bisa klik Review+create.

Microsoft Azure

Home

Create load balancer

Validation passed

Basics Frontend IP configuration Backend pools Inbound rules Outbound rules Tags Review + create

Basics

Subscription: Azure for Students
Resource group: MyLoadBalancer
Name: TA-demo-loadbalancer
Region: East Asia
SKU: Standard
Tier: Regional
Type: Public

Frontend IP configuration

Frontend IP configuration name: frontend-ip
Frontend IP configuration IP address: To be created

Backend pools

Backend pool name: TA-bp-loadbalancer

Inbound rules

Load balancing rule name: TA-lb-rule
Health probe name: TA-hp-lb

Outbound rules

None

Tags

None

Create < Previous Next > Download a template for automation Give feedback

8. Berikut tampilan ketika *Load Balancer* sudah berhasil terdeploy, klik copy *frontend IP address* untuk mengetes *Load Balancer* di dalam browser

Microsoft Azure

Home > CreateLoadBalancerBladeV2-20250517124342 | Overview

TA-demo-loadbalancer

Search resources, services, and docs (5/17)

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Automation

Help

Essentials

Resource group (move): MyLoadBalancer

Location (move): East Asia

Subscription (move): Azure for Students

Subscription ID: 67a6b8d1-15e1-4c56-8140-5aa42537ad1e

SKU: Standard

Tier: Regional

Tags (edit): Add tags

Backend pool: TA-bp-loadbalancer (2 virtual machines)

Load balancing rule: TA-lb-rule (TCP/80)

Health probe: TA-hp-lb (HTTP/80)

Inbound NAT rules: -

Outbound rules: -

Frontend IP address: 10.239.115.182 (TA-demo-lb-lb)

Configure high availability and scalability for your applications

Create highly-available and scalable applications in minutes by using built-in load balancing for cloud services and virtual machines. Azure Load Balancer supports TCP/UDP-based protocols and protocols used for real-time voice and video messaging applications. [Learn more](#)

Balance IPv4 and IPv6 addresses

Native dual-stack endpoints help meet regulatory requirements and address the fast-growing number of devices in mobile and IoT. [Learn more](#)

View frontend IP configuration

View backend pools

Build highly reliable applications

Load Balancer improves application uptime by routing traffic to healthy nodes. [Learn more](#)

View health probes

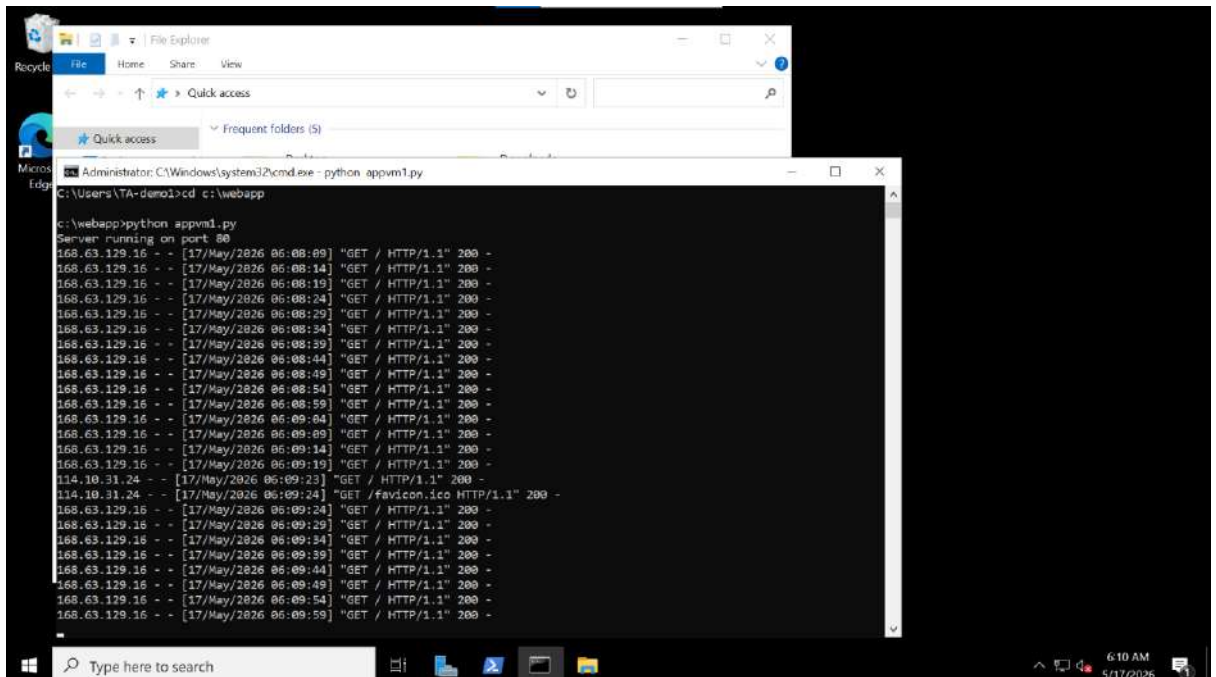
View load balancing rules

Secure your networks

Control network traffic and protect private networks using built-in network address translation (NAT). [Learn more](#)

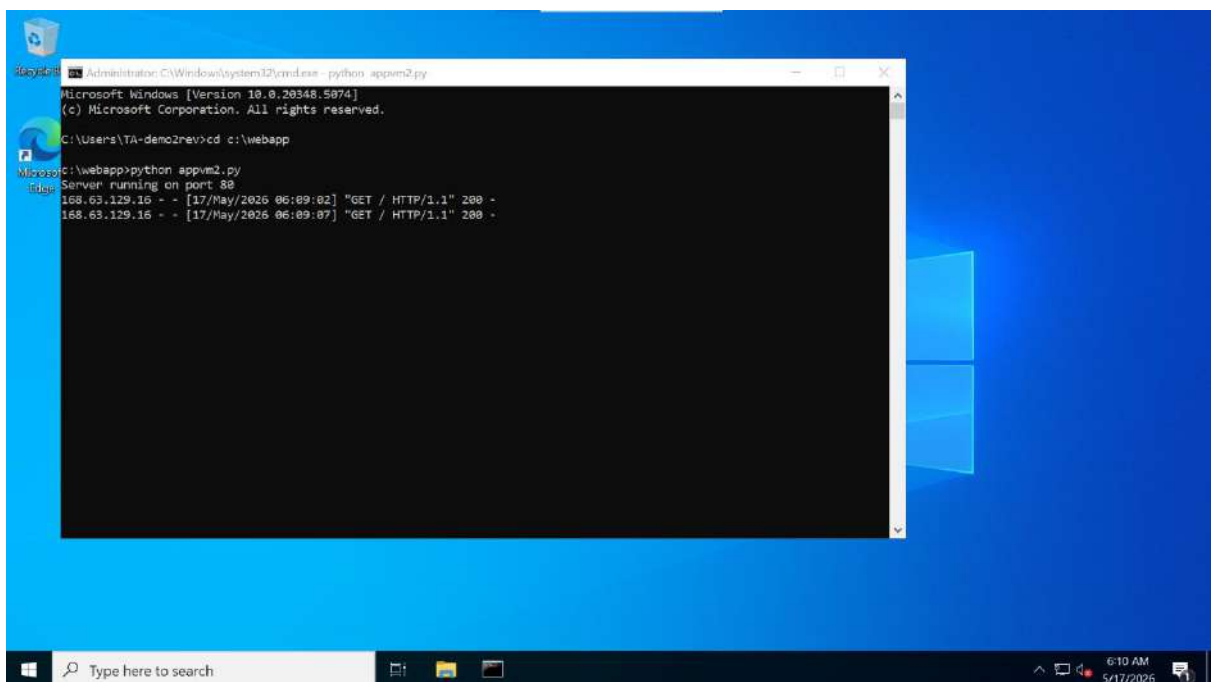
View inbound NAT rules

9. Selanjutnya sebelum mengetes Load Balancer berjalan atau tidak, pastikan python file pada VM1 dan VM2 sedang berjalan, seperti berikut ini:



```
Administrator: C:\Windows\system32\cmd.exe - python appvm1.py
C:\Users\TA-demo1>cd c:\webapp

c:\webapp>python appvm1.py
Server running on port 80
168.63.129.16 - - [17/May/2026 06:08:09] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:08:14] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:08:19] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:08:24] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:08:29] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:08:34] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:08:39] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:08:44] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:08:49] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:08:54] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:08:59] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:09:04] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:09:09] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:09:14] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:09:19] "GET / HTTP/1.1" 200 -
114.10.31.24 - - [17/May/2026 06:09:23] "GET / HTTP/1.1" 200 -
114.10.31.24 - - [17/May/2026 06:09:24] "GET /favicon.ico HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:09:24] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:09:29] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:09:34] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:09:39] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:09:44] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:09:49] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:09:54] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:09:59] "GET / HTTP/1.1" 200 -
```

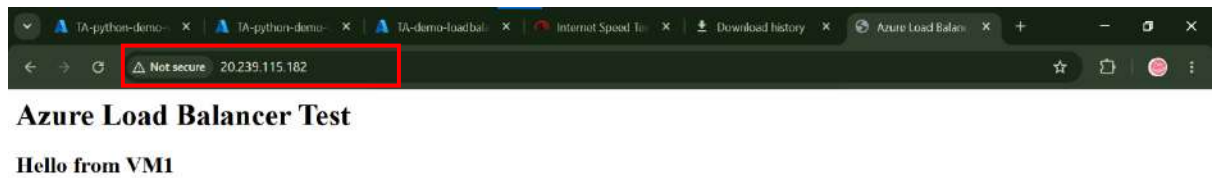


```
Administrator: C:\Windows\system32\cmd.exe - python appvm2.py
Microsoft Windows [Version 10.0.20348.5074]
(c) Microsoft Corporation. All rights reserved.

C:\Users\TA-demo2>cd c:\webapp

c:\webapp>python appvm2.py
Server running on port 80
168.63.129.16 - - [17/May/2026 06:09:02] "GET / HTTP/1.1" 200 -
168.63.129.16 - - [17/May/2026 06:09:07] "GET / HTTP/1.1" 200 -
```

10. Sekarang, saatnya mengetes Load Balancer di dalam browser, copy paste Public IP Address load balancer (lihat tahapan nomor 8). Berikut saat melakukan test di Chrome Browser, bisa dilihat bahwa di sini yang muncul adalah python file dari VM1. Perhatikan juga IP yang digunakan di sini.



11. Lalu, coba akses menggunakan browser lain, di sini menggunakan Firefox. Dengan IP yang sama, pada browser ini yang muncul adalah python file dari VM2, di sini menandakan bahwasannya Load Balancer berjalan dengan baik. Dengan menggunakan IP yang sama (20.239.115.182) load balancer mengatur traffic berpindah pindah dari VM1 dan VM2.

